Effective sample size is a variance statistic being read as a bias warning

The routine population-adjustment diagnostic panel scored as a classifier of realized error, answering catalog problem DIA-03

Ahmad Sofi-Mahmudi

2026-07-27

Abstract

Every population-adjusted indirect comparison is reported with a panel of numbers beside it: the effective sample size after weighting, the same figure as a percentage, the largest individual weight, standardized differences before and after adjustment. An appraisal committee reads them as evidence about whether to believe the estimate, and a widely quoted rule says to worry once absolute effective sample size falls below about 30 to 35. An audit of the open-problem catalog found that no published work has scored any of these as a classifier of realized error, with discrimination and calibration, against a known truth.

We do that. The design is an anchored indirect comparison with a continuous outcome and an identity link, chosen because on that scale the realized error of a linear estimator splits **exactly** into three pieces: chance covariate imbalance between the source arms, transport error, and outcome noise. Transport error is the only piece a covariate diagnostic could know about before an outcome is seen, so a diagnostic can be scored against the part it has a claim on rather than only against the part it does not. The design has 128 cells crossing an omitted effect modifier, a cross-moment modifier that no baseline table could ever reveal, overlap, source size, how correlated the omitted modifier is with the adjustment set, and a target dispersion ratio that destroys effective sample size while adding no bias wherever the modifier structure is correctly specified. 4,000 replicates per cell, four estimators.

The prespecified verdict is **the panel does not classify realized error at the thresholds in use**. Reading the panel is not reading a warning about bias: effective sample size separates the noise component at AUROC 0.847 and the transport component at 0.653. In the stratum where an effect modifier is omitted and the omission is diagnosable in principle, the ESS < 35 rule reaches sensitivity 0.309 at specificity 0.926. Three of the panel's members are one statistic written three ways, because $\mathrm{ESS}/n = 1/(1 + \mathrm{CV}^2(w))$ exactly and area under the ROC curve is invariant to monotone transformation; a fourth, the post-weighting balance on the matched moments, is zero at the solution of the calibration equations and cannot discriminate anything. Converting the residual imbalance on a measured-but-unadjusted covariate into the units of the estimand, using an interaction the source data can estimate, reaches 0.937 against the transport component where the panel reaches 0.851, and is blind exactly where the information is absent.

1 What the audit left open

The catalog entry DIA-03 originally said that whether population-adjustment diagnostics predict error "has never been quantified". Two independent auditors judged that too strong, and they were right. An ISPOR Europe 2024 simulation reported bias in unanchored MAIC once absolute effective sample size fell below roughly 30 to 35 (1), and Remiro-Azócar, Heath and Baio tied small effective sample size to underestimation of variability by the robust sandwich variance estimator (2). What survived the audit is narrower:

Neither treats a diagnostic as a classifier with discrimination and calibration against known truth.

One auditor added the objection that turns this from a complaint into a research question. Computing a diagnostic from the fitted data does not logically prevent it from predicting error. Sensitivity, specificity and calibration are defined only relative to a specified distribution of analyses and a prespecified threshold for material error; they are not intrinsic properties of a statistic. That is why the field has no numbers: nobody has written the distribution down. This study writes one down in advance, and reports everything twice, once under it and once with all cells weighted equally.

It also reports the primary results **within** strata of misspecification rather than over a mixture of them, which removes the dependence on our chosen frequencies entirely. That change came from an adversarial critique of this design, before the run.

2 What a covariate diagnostic could possibly know

Write the source trial's outcome as $Y_i = f(X_i) + \mathbb{1}(A_i)\{d_A + g_A(X_i)\} + \varepsilon_i$ with $\varepsilon_i \sim N(0, \sigma^2)$, and let the estimand be the transported effect in the target superpopulation, $\theta_{AC}(T) = d_A + \mathbb{E}_T\{g_A(X)\}$. The identity link does not make the heterogeneous conditional effect equal to the marginal one; what it buys is the absence of non-collapsibility, so the marginal effect is exactly the average of the conditional effects over the target covariate distribution.

MAIC, STC and the unadjusted comparison are all **linear in the outcome vector** for a fixed design: each is $\hat{\theta} = L(Y)$ for some linear functional L. For MAIC, $L(y) = \sum_{i\in A} w_i y_i / \sum_{i\in A} w_i - \sum_{i\in C} w_i y_i / \sum_{i\in C} w_i$; for STC, $L(y) = a^\top y$ with $a = M(M^\top M)^{-1} e_A$. Applying the same L to each piece of $\mathbb{E}[Y \mid X, A]$ gives an identity, not an approximation:

$$\underbrace{\hat{\theta} - \theta_{AC}(T)}_{\text{realized error}} = \underbrace{L(f)}_{\text{arm imbalance}} + \underbrace{L(\mathbb{1}(A)\{d_A + g_A\}) - \theta_{AC}(T)}_{\text{transport error}} + \underbrace{L(Y) - L(\mathbb{E}[Y \mid X, A])}_{\text{outcome noise}}.$$

Three consequences organize the whole study.

Only the middle term is what adjustment exists to remove. The first is chance covariate imbalance between the source arms, which randomization makes mean-zero but not zero in any one trial. The third is sampling error in the outcomes.

Only the middle term is knowable from covariates. The first and third depend on the realized outcomes; no function of X, the weights and a published baseline table has any information about ε. A diagnostic that fails to predict total realized error may simply be being asked to predict noise. A diagnostic that fails to predict the transport component has no such excuse. Both are reported, and the difference between them is the finding.

The identity holds per replicate. We verified it numerically to 3×10^{-14} before the run.

3 Two facts that algebra settles before any data exist

The panel is smaller than it appears.

Effective sample size, effective sample size as a percentage, and the coefficient of variation of the weights are one statistic. With $\bar{w}$ the mean weight and s^2 the population variance of the weights,

$$\frac{\text{ESS}}{n} = \frac{\left(\sum_i w_i\right)^2}{n \sum_i w_i^2} = \frac{n^2 \bar{w}^2}{n \cdot n\,(s^2 + \bar{w}^2)} = \frac{\bar{w}^2}{s^2 + \bar{w}^2} = \frac{1}{1 + \text{CV}^2(w)}.$$

Area under the ROC curve depends on the score only through its ranks, so a strictly monotone transformation cannot change it. Within a fixed source size the three therefore have **identical** discrimination as a matter of arithmetic. Across sizes they differ, because n varies. A submission that reports all three is reporting one number three times and calling it triangulation.

The post-weighting standardized difference on the matched moments is zero. The MAIC calibration equations are $\sum_i w_i\{h(X_i)-m_T\}=0$; the balance statistic is that expression divided by a standard deviation. It is set to zero by the thing being reported as evidence that it worked. In this run it never exceeded 1.4e-14.

Both are checked in the results rather than assumed, because the check is a test of the implementation.

4 Design

The full protocol was registered before the run. What matters for reading the results:

Four estimators. MAIC calibrating the target's means and standard deviations of the adjustment set; MAIC calibrating means only; STC with a heteroskedasticity-consistent variance (3); and no adjustment at all. All adjust for the same three covariates, so when a modifier is omitted they are misspecified in the same way. MAIC's variance is the M-estimation sandwich over the stacked calibration and arm-mean equations (4).

Three bias channels, differing in whether anything could see them. An omitted effect modifier X_4, which the source data can estimate an interaction for and a baseline table may report a mean for, so it is diagnosable in principle. A cross-moment term in X_1X_2 with source and target correlations of 0.30 and 0.70, which no marginal balance statistic can see and which no baseline table reports. And extrapolation, which moves effective sample size and bias together and is what gives effective sample size its only real chance.

The factor that makes the study falsifiable. The target's covariate standard deviations are either equal to the source's or 25% larger. Matching second moments to a more dispersed target destroys effective sample size while adding no bias in the well-specified stratum, because the modifier is linear in the adjustment set and its mean is matched either way. Without it, every low-effective-sample-size cell would also be a poor-overlap cell, and a variance statistic would look like a bias statistic because the design never separated the two.

The reference. Material error means $|\hat{\theta}_{AC}-\theta_{AC}(T)|>0.20$, a fifth of the outcome standard deviation. No diagnostic enters its definition.

Scale. 128 cells, 4,000 replicates each, four estimators. MAIC found a solution on 99.3% of replicates; the rest are reported per cell rather than dropped silently.

5 Results

5.1 The panel at the thresholds in use

Table 1: Sensitivity and specificity for material error, within each misspecification stratum, cells weighted equally inside a stratum. Monte Carlo standard errors in brackets.

diagnostic (sensitivity / specificity)	well.specified	omitted.modifier	cross.moment	both
ESS	0.326 / 0.923	0.309 / 0.926	0.265 / 0.918	0.251 / 0.915
ESS %	0.742 / 0.623	0.735 / 0.654	0.649 / 0.607	0.642 / 0.633
largest weight	0.350 / 0.919	0.332 / 0.920	0.285 / 0.913	0.269 / 0.909
balance, matched moments	0.000 / 1.000	0.000 / 1.000	0.000 / 1.000	0.000 / 1.000
imbalance before weighting	0.822 / 0.408	0.823 / 0.431	0.767 / 0.403	0.771 / 0.440
Mahalanobis distance	0.656 / 0.692	0.652 / 0.721	0.570 / 0.683	0.564 / 0.707
balance, unmatched covariate	0.559 / 0.610	0.571 / 0.639	0.520 / 0.608	0.543 / 0.681
estimated bias (proposed)	0.099 / 0.962	0.393 / 0.807	0.091 / 0.957	0.358 / 0.827

The prespecified bar was sensitivity at least 0.80 with specificity at least 0.50 in every stratum, lower Monte Carlo limits above both. No rule met it.

In the omitted-modifier stratum, which is where a diagnostic has both something to find and the information with which to find it, the published cutoff reaches sensitivity 0.309 (0.001) at specificity 0.926. Two members of the panel reach high sensitivity by firing on nearly everything: the pre-weighting imbalance rule fires on 70.6% of analyses in that stratum and the unmatched-balance rule on 47.4%. A rule that warns about almost every analysis is not a classifier, and its specificity says so: 0.431 and 0.639.

Replicates on which MAIC had no solution: 3,747. Bracketing the primary rule's sensitivity by counting every one of them as a caught failure, then as a missed one, gives 0.285 and 0.273.

5.2 The statistics carry information the thresholds do not use

Table 2: Discrimination under the declared deployment distribution. The last two columns are the point of the study: which component of the realized error does each statistic actually track?

	diagnostic	AUROC	AUPRC	vs outcome noise	vs transport error
ess	ESS	0.731 (0.001)	0.707	0.847	0.653
ess_pct	ESS %	0.723 (0.001)	0.698	0.813	0.667
cv_w	CV(w)	0.723 (0.001)	0.698	0.813	0.667
max_w	largest weight	0.741 (0.001)	0.716	0.833	0.660
smd_matched	balance, matched moments	0.550 (0.001)	0.509	0.571	0.538
smd_pre	imbalance before weighting	0.683 (0.001)	0.671	0.799	0.655
maha	Mahalanobis distance	0.682 (0.001)	0.669	0.796	0.656
smd_unmatched	balance, unmatched covariate	0.606 (0.001)	0.592	0.663	0.665
bias_hat	estimated bias (proposed)	0.604 (0.001)	0.591	0.643	0.684
orc_cross	oracle: cross-moment	0.581 (0.001)	0.551	0.505	0.809

Discrimination is not zero. Nothing here is uninformative in the sense of being independent of error. But the ordering in the last two columns is the answer to what the panel is measuring: every weight-based statistic reads the noise channel better than the transport channel, and the gap for effective sample size is 0.194. That is the sense in which a variance statistic is being read as a bias warning.

The oracle row separates two different failures. The cross-moment oracle uses the target's true $\mathbb{E}[X_1 X_2]$, which no baseline table reports, and reaches 0.809 against transport error while everything computable sits near 0.667. Where the oracle wins, the information is missing rather than the statistic being the wrong function of it.

One row needs a warning label. The balance statistic on the matched moments shows AUROC 0.550, which is not 0.5 and might be read as weak signal. It is not. That statistic never exceeded 1.4e-14 anywhere in the run: what is being ranked is floating-point residual from the calibration solver, and the residual is larger in cells where the equations were harder to solve, which are also harder cells. The apparent discrimination is discrimination of arithmetic difficulty. Scoring a statistic that is identically zero produces a number, and the number means nothing.

5.3 The panel is smaller than it looks

Table 3: The algebraic identities, checked. Within a fixed source size the three weight-dispersion statistics have the same AUROC to machine precision, and the matched-moment balance statistic is zero.

source per arm	ESS	ESS %	CV(w)	max matched imbalance
150	0.706	0.706	0.706	1.3e-14
400	0.740	0.740	0.740	1.4e-14

5.4 Calibration

A diagnostic is calibrated if a mapping from it to a risk of material error, fitted once and locked, tells the truth on analyses it has not seen. The mapping here is fitted on odd-numbered design cells and evaluated on even-numbered ones, which is the question an analyst faces; refitting within the same cells is reported as the optimistic bound (5).

Table 4: Calibration of a locked logistic mapping from each diagnostic to the risk of material error. Perfect calibration is intercept 0 and slope 1.

	diagnostic	intercept	slope	calibration error, new cells	calibration error, same cells
ess	ESS	0.297	0.888	0.065	0.066
ess_pct	ESS %	0.248	0.837	0.061	0.069
max_w	largest weight	0.243	0.801	0.069	0.091
smd_pre	imbalance before weighting	0.648	0.845	0.143	0.075
maha	Mahalanobis distance	0.649	0.844	0.143	0.075
smd_unmatched	balance, unmatched covariate	0.534	0.765	0.127	0.101
bias_hat	estimated bias (proposed)	0.535	0.766	0.130	0.086

The weight-dispersion statistics calibrate tolerably across cells they have not seen: effective sample size has intercept 0.297 and slope 0.888, with an integrated calibration error of 0.065. The overlap statistics do not: the pre-weighting imbalance mapping carries an error of 0.143 on new cells against 0.075 when refitted on the cells it is scored in, so nearly half of its apparent calibration is memory of the setting. That gap is the reason the split is across cells rather than across replicates, and an adversarial review of the design is the reason it is.

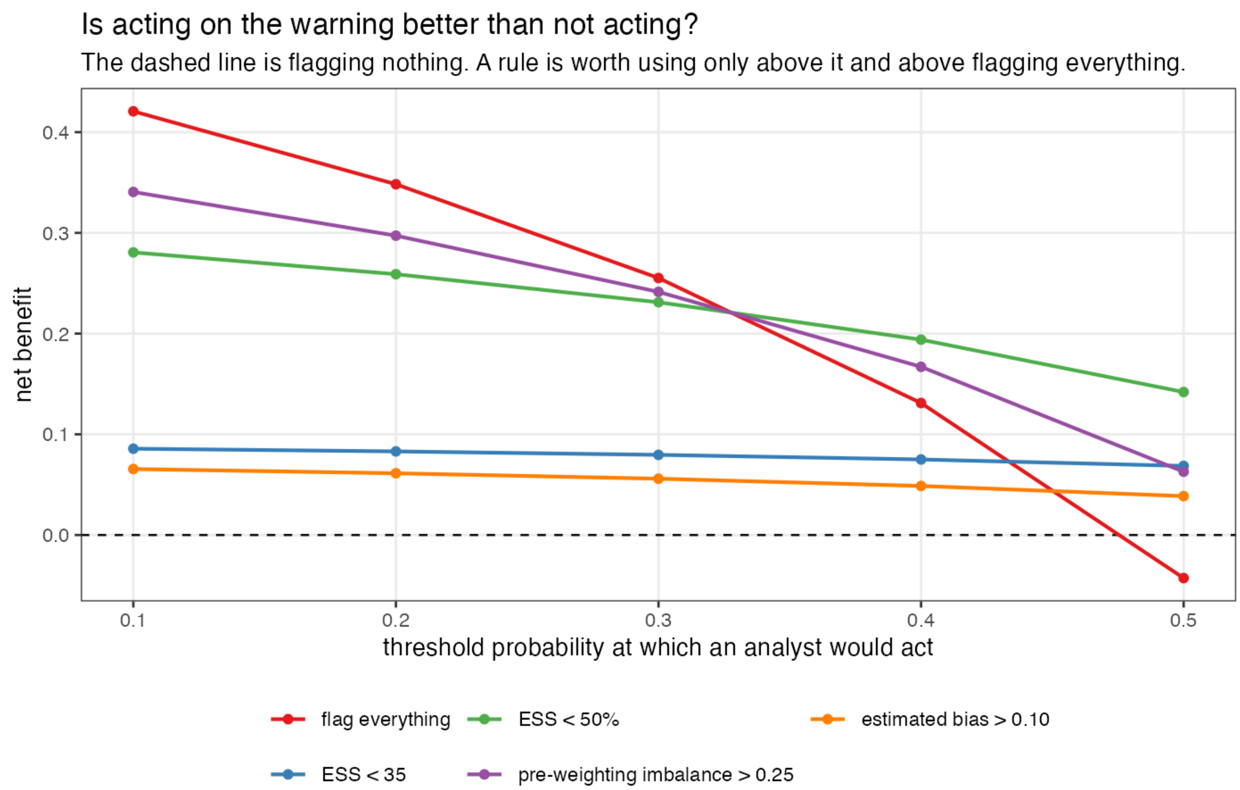
5.5 Is acting on the warning better than not acting?

Net benefit relative to flagging nothing, over a range of threshold probabilities at which an analyst would act (6). A rule earns its place only above both flagging nothing and flagging everything.

Table 5: Net benefit relative to flagging nothing, under the deployment weights.

action threshold	flag nothing	flag everything	ESS < 35	ESS < 50%	imbalance > 0.25	Mahalanobis > 1	estimated bias > 0.10
0.1	0	0.421	0.086	0.281	0.341	0.242	0.066
0.2	0	0.348	0.083	0.259	0.297	0.224	0.061
0.3	0	0.255	0.080	0.231	0.241	0.202	0.056
0.4	0	0.131	0.075	0.194	0.167	0.172	0.049
0.5	0	-0.043	0.069	0.142	0.063	0.130	0.039

Every rule in the panel is beaten by flagging everything at action thresholds up to 0.300. An analyst who would act on a 10.0% chance of material error does better scrutinizing every population adjustment than reading any of these numbers. Only once the action threshold reaches 0.400, meaning the analyst will only intervene when material error is at least that likely, does a diagnostic earn its place, and the one that does it is the relative effective-sample-size rule rather than the absolute one that gets quoted.



5.6 The four prespecified mechanism claims

Table 6: The four claims registered before the run, and whether the run supports them.

claim	holds	evidence
It is a variance statistic, not a bias statistic	yes	Effective sample size discriminates outcome noise at AUROC 0.847 and transport error at 0.653, a gap of 0.194 against a prespecified 0.10.
The threshold is not transportable	no	Across the 24 cells the rule flags on most replicates, the material-error rate runs from 0.751 to 0.902, a span of 0.150 against a prespecified 0.30. Across the 88 cells it is silent on, the rate runs from 0.045 to 0.835.
The panel is one statistic	yes	Within a fixed source size the three agree to 0.0e+00 in AUROC, and the matched-moment balance statistic never exceeds 1.4e-14.
A better statistic is available from the same information	no	In the diagnosable stratum, against the transport component, the proposal reaches 0.937 and the best routinely reported diagnostic 0.851, a gain of 0.086 against a prespecified 0.10.

Two of the four hold and two do not, and the two that do not are reported as they were registered rather than adjusted until they did.

The threshold-transportability claim failed as written, and the complementary quantity is the interesting one. We registered a span of at least 0.30 in the material-error rate across the cells the rule *flags*. The observed span is 0.150, and the reason is visible in Figure 1: the flagged cells are all in the low-effective-sample-size region where almost everything is wrong, so there is little room for a spread. The cells the rule is **silent** on are where a spread matters, because silence is what an analyst acts on, and there the material-error rate runs from 0.045 to 0.835, a span of 0.789 across 88 cells. That is a stronger result than the one registered and it was not prespecified, so it is reported as an observation and not as a passed test.

The proposal missed its bar too. Registered at a gain of at least 0.10 in area under the ROC curve against the transport component in the diagnosable stratum; observed 0.937 against 0.851, a gain of 0.086. It is a real improvement and it is not the one we said would count.

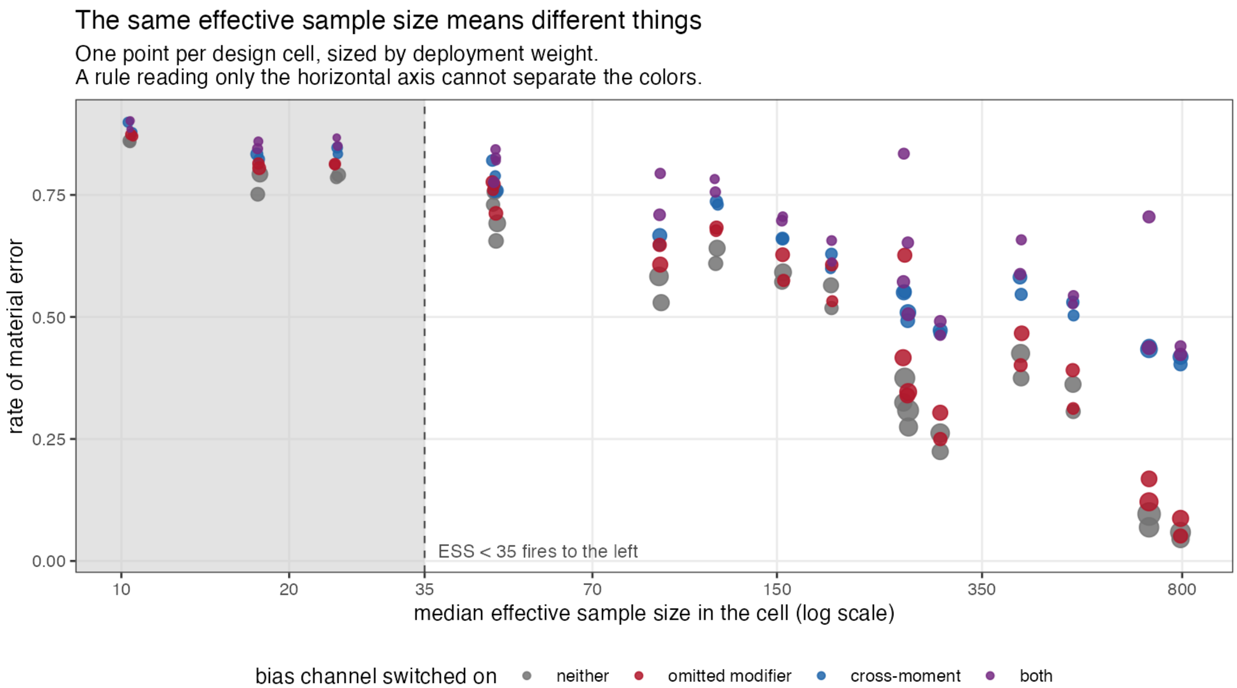
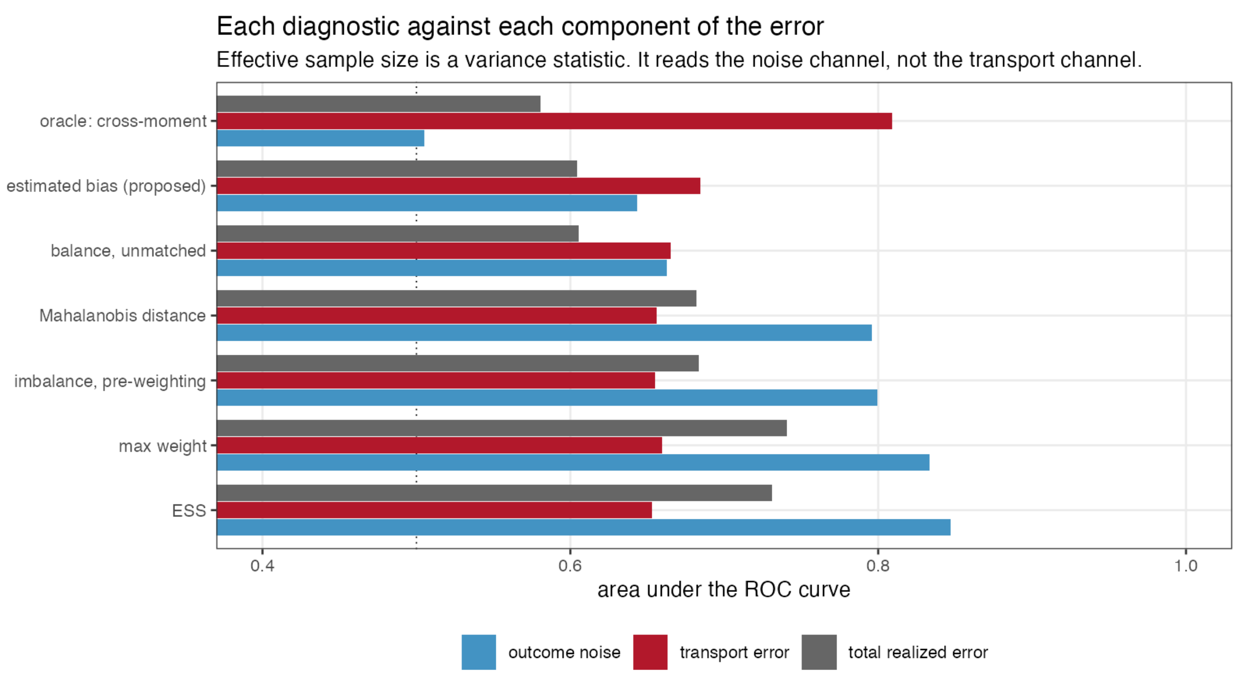


Figure 1



5.7 The other estimators

Table 7: The same replicates seen by the other three estimators. Median effective sample size is over the whole design.

method	diagnostic	AU-ROC	vs transport	material error	median ESS	cover-age
MAIC, means only	ESS	0.704	0.599	0.433	261.783	0.730
MAIC, means only	imbalance before weighting	0.685	0.607	0.433	261.783	0.730
MAIC, means only	Mahalanobis distance	0.683	0.608	0.433	261.783	0.730
MAIC, means only	balance, unmatched covariate	0.602	0.624	0.433	261.783	0.730
MAIC, means only	estimated bias (proposed)	0.602	0.641	0.433	261.783	0.730
STC	imbalance before weighting	0.656	0.635	0.340	550.000	0.533
STC	Mahalanobis distance	0.655	0.636	0.340	550.000	0.533
STC	balance, unmatched covariate	0.606	0.619	0.340	550.000	0.533
STC	estimated bias (proposed)	0.611	0.628	0.340	550.000	0.533
no adjustment	imbalance before weighting	0.804	0.891	0.596	550.000	0.357
no adjustment	Mahalanobis distance	0.804	0.891	0.596	550.000	0.357
no adjustment	balance, unmatched covariate	0.802	0.892	0.596	550.000	0.357
no adjustment	estimated bias (proposed)	0.798	0.873	0.596	550.000	0.357

The unadjusted comparison is the control that explains the rest of the paper. Its pre-weighting imbalance statistic predicts its own transport error at AUROC 0.891, against 0.655 for the same statistic applied to MAIC. The statistic is not weak. It is an excellent measure of the gap between the two populations, and the gap between the two populations is exactly the error of an estimator that does nothing about it. Once weighting has closed those gaps, the statistic is measuring something that has been removed, and what remains is the part of the modifier structure the weights never touched. The diagnostic is informative about the problem and uninformative about the solution.

Means-only MAIC is worth a line for the same reason. It carries a median effective sample size of 261.783 against 170.748 for the version that also matches standard deviations, and a lower material-error rate, 0.433 against 0.479. Calibrating second moments that the modifier structure does not use costs effective sample size and buys nothing here. A reader should not generalize that: with a modifier that is nonlinear in the covariates, those moments would matter.

6 What this answers, and what it does not

DIA-03. The entry asked for the routine panel to be scored as a classifier of realized error with discrimination and calibration against a known truth, under a specified scenario distribution and a prespecified material-error threshold. That is done, for the MAIC weighting panel and the overlap statistics it shares with STC. The second half of the entry, the mismatch between the hierarchical level at which cross-validation is reported and the level at which the prediction is made, is untouched: no PSIS-LOO is computed at any level here, and grouped study-level leave-one-out is no more implemented in this study than it is in the packages. That half remains open.

DIA-06. Answered in part, and the part is small. Four estimators from two families see the same replicates, so their failure modes can be compared on identical data; but ML-NMR, NMI, doubly robust estimators and flexible learners are absent, and the entry names five families. A design critique made this point before the run and it is accepted rather than argued with.

What the mechanism deliberately makes true. Randomization holds. The prognostic and modifier functions are shared across trials, so conditional constancy holds. Covariates are normal. The link is the identity. Target moments are exact, because study 1 of this program already measured what their sampling error costs and carrying it here would add a noise channel no source-side diagnostic can observe. Every estimator adjusts for the same covariates.

Therefore the study says nothing about non-collapsible effect measures, time-to-event outcomes, skewed or discrete covariates, unanchored comparisons, target-moment sampling error, or the diagnostics specific to Bayesian population adjustment. A binary or survival outcome would add a component of error that is a collapsibility artifact rather than an adjustment failure, and the exact decomposition in Section 2 would not be available; that is the price paid for the instrument, and it is the largest limitation here.

The deployment weights are a judgment, not a measurement. Nothing in this program estimated how often an effect modifier is omitted in practice. That is why the primary results are within strata, where the judgment does not enter, and why every mixture-level figure is reported under two weightings.

The proposal is ours. $\hat{b}$ was invented for this study, scored at a threshold fixed in advance, against the same reference as everything else, and its prespecified claim was restricted before the run to the stratum and the error component where it has any information at all. It is not a validated diagnostic and this study does not make it one; it is a demonstration that the information to do better is sometimes already in the analyst's hands.

7 Peer review

This manuscript was reviewed by three independent reviewers over two rounds. The reports, the authors' responses and the editorial decision are published in full and unedited alongside it.

8 Reproducibility

All code is in the study directory. `R/00-config.R` holds the design, the thresholds and the decision rule; `R/01-dgm.R` the mechanism; `R/02-estimators.R` the four estimators and the exact error decomposition; `R/03-diagnostics.R` the panel; `R/04-run.R` the run; `R/05-analyze.R` the analysis; `R/06-figures.R` the figures. Replicates are seeded from one master seed through independent L'Ecuyer-CMRG streams, so a replicate draws the same data whatever the core count. `results/provenance.md` records the R version and package versions that produced these numbers. Raw replicate output is regenerable from the seed and is not tracked; the tracked artifacts are the summary tables, the figures and the rendered manuscript.

References

1. MSR65 can low effective sample size in matching-adjusted indirect comparisons (MAICs) lead to bias? Findings from a simulation study [Internet]. ISPOR Europe 2024, abstract published in Value in Health; 2024. Available from: https://www.valueinhealthjournal.com/article/S1098-3015(24)05162-3/fulltext

2. Antonio Remiro-Azócar, Anna Heath, Gianluca Baio. Methods for population adjustment with limited access to individual patient data: A review and simulation study. Research Synthesis Methods. 2021;12(6):750–75. doi:10.1002/jrsm.1511

3. James G. MacKinnon, Halbert White. Some heteroskedasticity-consistent covariance matrix estimators with improved finite sample properties. Journal of Econometrics. 1985;29(3):305–25. doi:10.1016/0304-4076(85)90158-7

4. Leonard A. Stefanski, Dennis D. Boos. The calculus of m-estimation. The American Statistician. 2002;56(1):29–38. doi:10.1198/000313002753631330

5. Peter C. Austin, Ewout W. Steyerberg. The integrated calibration index (ICI) and related metrics for quantifying the calibration of logistic regression models. Statistics in Medicine. 2019;38(21):4051–65. doi:10.1002/sim.8281

6. Andrew J. Vickers, Elena B. Elkin. Decision curve analysis: A novel method for evaluating prediction models. Medical Decision Making. 2006;26(6):565–74. doi:10.1177/0272989X06295361